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FIN2SUM: Advancing AI-Driven Financial Text Summarization with LLMs

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***Abstract* – In the modern financial sector, the rapid digitalization of financial reports necessitates efficient and reliable text summarization tools. This research introduces FIN2SUM, a novel framework designed for summarizing the managerial analysis and discussion sections of 10-K reports from top NASDAQ-listed companies. The study aims to evaluate Large Language Models (LLMs) in financial text summarization, highlighting LLAMA-2's adeptness in processing complex financial information, thus making FIN2SUM a vital tool for analysts and decision-makers. The methodology includes a thorough evaluation of three state-of-the-art LLMs—LLAMA-2, FLAN, and Claude 2—using BERT and ROUGE scores. The research concludes that FIN2SUM, enhanced by LLAMA-2, significantly advances AI-driven financial text summarization.**

Keywords – Financial Text Summarization; Large Language Models (LLMs); AI in Financial Analysis; 10-K Reports Summarization; FIN2SUM Framework; LLAMA-2 and Financial Data; BERT and ROUGE Metrics; AI-Driven Decision Making; Financial Data Processing; Semantic Similarity Analysis; Financial Reporting Automation

I. INTRODUCTION

The advent of Generative AI represents a significant milestone in the field of artificial intelligence, particularly in the realm of Natural Language Processing (NLP). These advanced AI systems, known as Large Language Models (LLMs), have demonstrated remarkable capabilities in understanding and generating human-like text. Their application has been transformative across various domains, with financial document summarization being a notable area of impact. The ability of these models to distil lengthy and complex texts into concise, coherent summaries has opened up new possibilities for efficient data processing and analysis[1].

Within the financial sector, the analysis and summarization of comprehensive documents, such as 10-K reports, pose a significant challenge due to their volume and complexity. These reports contain crucial insights into a company's financial health and strategic direction, making their swift and accurate interpretation vital for stakeholders. The FIN2SUM framework, developed in this research, specifically addresses the need for efficient summarization of these reports. It utilizes the prowess of LLMs to transform dense financial narratives into concise summaries, thereby facilitating quicker and more effective decision-making processes[2].

The motivation for this research stems from the urgent need to manage the ever-increasing volume of financial data in an efficient manner. The central research question revolves around the efficacy of LLMs in accurately summarizing complex financial texts. The objective is to evaluate and compare the performance of several state-of-the-art LLMs—LLAMA-2, FLAN-T5-Base, and Claude 2—within the FIN2SUM framework[3]. The methodology involves a rigorous analysis using BERT SCORE[4] and ROUGE METRIC [5] to assess the semantic and structural fidelity of the summaries generated by these models[6]. This approach aims to establish a comprehensive understanding of the models' capabilities in handling financial texts[7].

The FIN2SUM framework embodies an innovative architecture that integrates multiple Large Language Models, each fine-tuned for financial text summarization. Key components include LLAMA-2, known for its ability to capture the nuances of financial narratives, and FLAN, acclaimed for its advanced text generation capabilities. The framework employs a dual-metric evaluation strategy, utilizing both BERT scores[8]for semantic similarity and ROUGE metrics[9] for n-gram overlap analysis. This combination ensures a holistic evaluation of the models' performance in generating summaries that are not only precise but also coherent and contextually relevant. The inclusion of a user-friendly interface enhances the accessibility and practicality of the framework, making it a potent tool for financial analysts and decision-makers.

II. LITERATURE REVIEW

A. Evolution and Impact of Large Language Models (LLMs) in Finance

The introduction of Large Language Models (LLMs) into financial text analysis has marked a transformative shift in the field. Models like GPT-3 have pioneered in-context learning (ICL), enabling them to generate insightful narratives from a range of financial data, including news articles, earnings call transcripts, and social media posts. This advancement has been particularly significant in the financial sector, which is characterized by its complex and voluminous textual data. In response, domain-specific LLMs have emerged, such as BloombergGPT. This model, boasting 50 billion parameters and trained on a mix of financial and general data, has demonstrated exceptional performance in financial tasks while also maintaining robust capabilities in general-purpose benchmarks[10][11].

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B. Specific Models and Their Contributions

BloombergGPT*:* It is a prime example of a domain-specific LLM with a dataset comprising financial news, filings, press rel4eases, and web-scraped documents. This model outperforms existing models on financial tasks without compromising on general LLM benchmarks[12].

FinBERT*:* This model was developed by pretraining a financial domain-specific BERT on a large financial communication corpora. It has shown superior performance in financial sentiment classification tasks over generic domain BERT models[13].

FinGPT*:* An open-source model aiming to democratize Internet-scale financial data. It addresses the need for accessible financial LLMs with a focus on automatic data curation and lightweight, low-rank adaptation techniques[14].

GPT-FinRE*:* A model employing GPT-3.5 Turbo and GPT-4 for financial relation extraction. It utilizes a learning-free dense retriever and a learning-based retriever, achieving a notable F1-score of 0.718 on the REFinD dataset[15].

C. Challenges and Opportunities in Financial Text Summarization

Financial texts such as 10-K reports and earnings calls are characterized by their extensive volume and complexity, presenting a significant challenge for efficient summarization. Extracting valuable insights from these documents is crucial for financial analysis and decision-making. The introduction of in-context learning (ICL) in Large Language Models (LLMs) offers a promising solution. ICL enables LLMs to adapt to the specific nuances of the financial domain through context-specific demonstrations, reducing the need for extensive retraining of the models[16]. Additionally, the financial landscape is constantly evolving, marked by its time-sensitive and dynamic nature. This requires models that can quickly adapt to new information and rapidly changing market conditions, ensuring that the financial insights derived remain relevant and accurate[17].

D. Future Directions and Potential Applications

Future advancements in the field of financial text analysis could center around developing adaptive and customizable models. These models would be capable of being easily fine-tuned for specific financial tasks, capitalizing on the inherent labeling capabilities of the market. The potential applications of such models are vast, ranging from robo-advising and quantitative trading to portfolio optimization and risk management[14]. This broadens the spectrum of their utility in the financial sector. Additionally, there is scope for enhancing relation extraction in financial texts. By leveraging advanced GPT models combined with innovative retrieval mechanisms, the accuracy and efficiency of analyzing financial documents could be significantly improved. This would provide deeper insights and more effective data utilization in financial decision-making processes[18].

III. PHILOSOPHICAL UNDERPINNINGS

This study explores the integration of AI and Large Language Models (LLMs) in financial analysis, focusing on transforming complex financial data into accessible insights. It delves into the epistemological aspects of AI in finance, examining LLMs' role in data summarization, ethical AI use, and knowledge democratization.

1. ***Epistemology and AI:*** Integrates AI with epistemology, leveraging LLMs to transform the understanding of complex financial data[19].
2. ***Understanding via Summarization:*** Utilizes summarization to make complex financial knowledge more accessible and manageable[20].
3. ***Ontology of Financial Data:*** Recognizes financial data as a reflection of a company's health and strategy, not just numbers and text[21].
4. ***AI as an Insight Tool:*** Employs AI to reveal deeper truths and patterns in financial data[21].
5. ***Ethical AI Use:*** Focuses on unbiased and transparent AI processing for ethical decision-making in finance[22].
6. ***Democratization of Knowledge:*** Aims to make intricate financial reports understandable to a wider audience, enhancing financial literacy[23].
7. ***Enhanced Decision-Making:*** Contributes to better decision-making by providing clearer and more concise financial information[24].
8. ***Methodological Rigor:*** Adopts a scientific, evidence-based approach to evaluate LLMs in financial text summarization[25].

In summary, our research highlights the efficacy of LLMs in financial text summarization, emphasizing ethical AI use and transparency. It contributes to financial literacy by making intricate financial data comprehensible to a broader audience, thereby enhancing decision-making in the financial sector.

IV. METHODOLOGY

The study's methodology encompasses data pre-processing of managerial sections from NASDAQ-listed companies' 10-K reports, focusing on 2022 publications. We employed advanced LLMs like LLAMA-2, FLAN, and Claude 2 for text summarization, augmented by text chunking, word vector implementation, and key financial relationship identification using these models.

A. Data Pre-Processing

- Source: Data comprises managerial analysis and discussion sections from 10-K reports of the top 100 NASDAQ-listed companies.
- Sourcing Method: Extracted from official online sources and Bloomberg terminals.
- Sample Size and Duration: Focus on reports published in 2022.

B. Text Summarization

- LLMs Used: LLAMA-2, FLAN, and Claude 2.
- Architecture: Each LLM represents advanced NLP capabilities.

C. Text Chunking

- Approach: Divide texts into smaller segments for efficient LLM processing.

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D. Word Vector Implementation

- Process: Convert text data into numerical format for LLM processing.

E. Relation Identifying

- Technique: LLMs identify key financial relationships within texts.

F. User Prompt Integration

- Functionality: Allows user-specified summarization requirements.

G. Evaluation Metrics

- ROUGE Metrics: Measures structural fidelity via n-gram overlap.
- BERT Score: Assesses semantic similarity between original texts and summaries.

H. Evaluation Process

- Method: Comparative analysis of LLMs using BERT and ROUGE scores.

For evaluation, we utilized ROUGE metrics for structural fidelity and BERT scores for semantic similarity. The methodology involved a comparative analysis of the LLMs, integrating user-specific summarization prompts. This approach ensures precise and effective financial text analysis, aligning with contemporary AI standards.

V. RESULT ANALYSIS

In our evaluation of different Large Language Models (LLMs) for text summarization, we focused on two key metrics: ROUGE (Reflecting overlap and coherence) and BERT (Measuring semantic similarity). This analysis compares the performance of LLAMA2-7B, FLAN-T5-Base, and Claudi AI across these metrics to determine their effectiveness in natural language understanding and summarization tasks.

A. ROGUE METRICS (Rouge-1, Rouge-2, and Rouge-L Metrics):

In the analysis, LLAMA2-7B distinctly outperforms FLAN-T5-Base and Claudi AI in terms of ROUGE metrics. It demonstrates superior proficiency in unigram and bigram overlaps (ROUGE-1, ROUGE-2) as well as in capturing the longest common subsequence's (ROUGE-L), signifying a stronger alignment with reference content. In comparison, while FLAN-T5-Base and Claudi AI exhibit comparable capabilities in ROUGE-2 and ROUGE-L, Claudi AI slightly surpasses in ROUGE-1, indicating a marginally enhanced ability in detecting unigram overlaps.

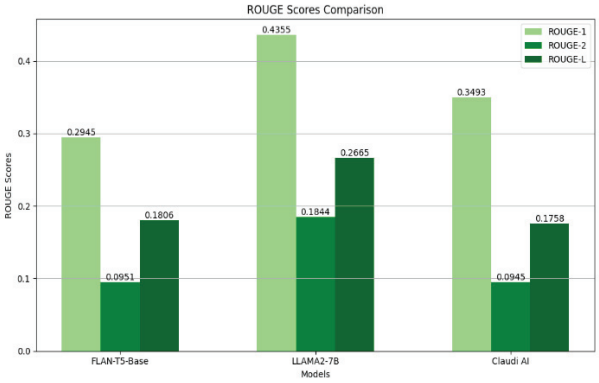


Fig. 1. Bar graph of ROGUE Metrics.

B. BERT Score Metrics (Precision, Recall, F1):

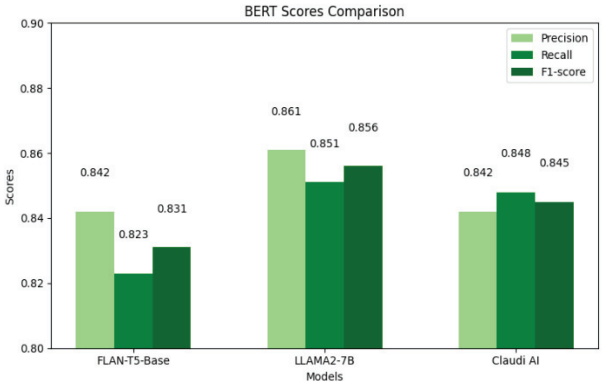


Fig. 2. :Bar graph of BERT Score.

The BERT score analysis reveals relatively similar performances among the models, suggesting that each has a robust capability in maintaining semantic coherence and relevance. LLAMA2-7B marginally leads, especially in Recall and F1 scores, indicating its slightly superior ability to retain relevant content and provide aS balanced performance in semantic similarity tasks.

Overall, **LLAMA2-7B** appears to be the most effective model across the metrics used, with strengths in both lexical overlap (ROUGE metrics) and semantic similarity (BERT Scores). This suggests its potential for applications requiring high-quality natural language understanding and generation. The closely matched performances in BERT Scores across all models also highlight the advancements in LLMs in maintaining semantic coherence, an essential aspect of natural language processing tasks.

TABLE I. PERFORMANCE OF DIFFERENT FIN2SUM MODELS ON THREE LARGE LANGUAGE MODEL

LLM Models	ROUGE-1	ROUGE-2	ROUGE-L	BERT Score (P/R/F1)
FLAN-T5-Base	0.2945	0.0951	0.1806	0.842/0.823/0.831
LLAMA2-7B	0.4355	0.1844	0.2665	0.861/0.851/0.856
Claudi AI	0.3493	0.0945	0.1758	0.842/0.848/0.845

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VI. ARCHITECTURE

A. Data Source and Collection

- ***Source:*** Financial 10-K Statements from NASDAQ-listed companies. Collection ***Method:*** Systematic gathering from SEC's EDGAR database.
- ***Formats:*** Inclusion of PDF, text, and XML formats.
- ***Metadata:*** Detailed recording of company names and report publication dates.
- ***Duration:*** The data covers reports from 2022. Sample Size: 100 companies.

B. Data Engineering

- ***Extraction:*** Tailored process for text extraction from PDFs. Structuring: Conversion of text into a machine-readable format. Normalization and Preprocessing: Standardization for AI model interpretability.
- ***Quality Assurance:*** Rigorous checks for data accuracy and reliability.

C. Data Processing with LLMs

- ***LLMs Used:*** LLMA-2, Google FLAN, Claude AI, GPT-3.5. Components: Transformer architecture [15], embeddings, and other internal mechanisms of LLMs [16]. ***Initialization:*** Fine-tuning for financial text nuances.
- ***Interface:*** User-friendly query interfaces. ***Model Selection:*** Strategic use of single or ensemble models for optimal summarization. ***Focus:*** Enhancing generative capabilities through fine-tuning for financial reporting.

D. Application Layer

User Interface: Development of a web/app-based platform. Backend Server: Establishment of a robust server for request handling.

Feedback Mechanism: Integration for user experience and summary quality improvement.

Data Security: Emphasis on encryption and compliance with FFIEC and GDPR for financial data protection.

VII. CONCLUSION, FINDINGS & FUTURE WORK

A. Conclusion

The study introduces a significant improvement in AI-driven financial text summarization with FIN2SUM. Using advanced Large Language Models like LLAMA-2, we've developed a method to efficiently summarize complex financial reports, particularly 10-K reports from leading NASDAQ companies. This approach improves accuracy and efficiency in financial analysis and opens new avenues for AI applications in finance.

B. Findings

Model Performance: LLAMA-2 outperformed other models like FLAN and Claude 2 in summarizing financial texts, showing better understanding and coherence.

Challenges: We identified issues like complex financial terminology and the need for contextual knowledge. LLMs are promising in addressing these.

Integrating Philosophy and AI: We've combined philosophical concepts with AI for ethical use and wider access to financial information.

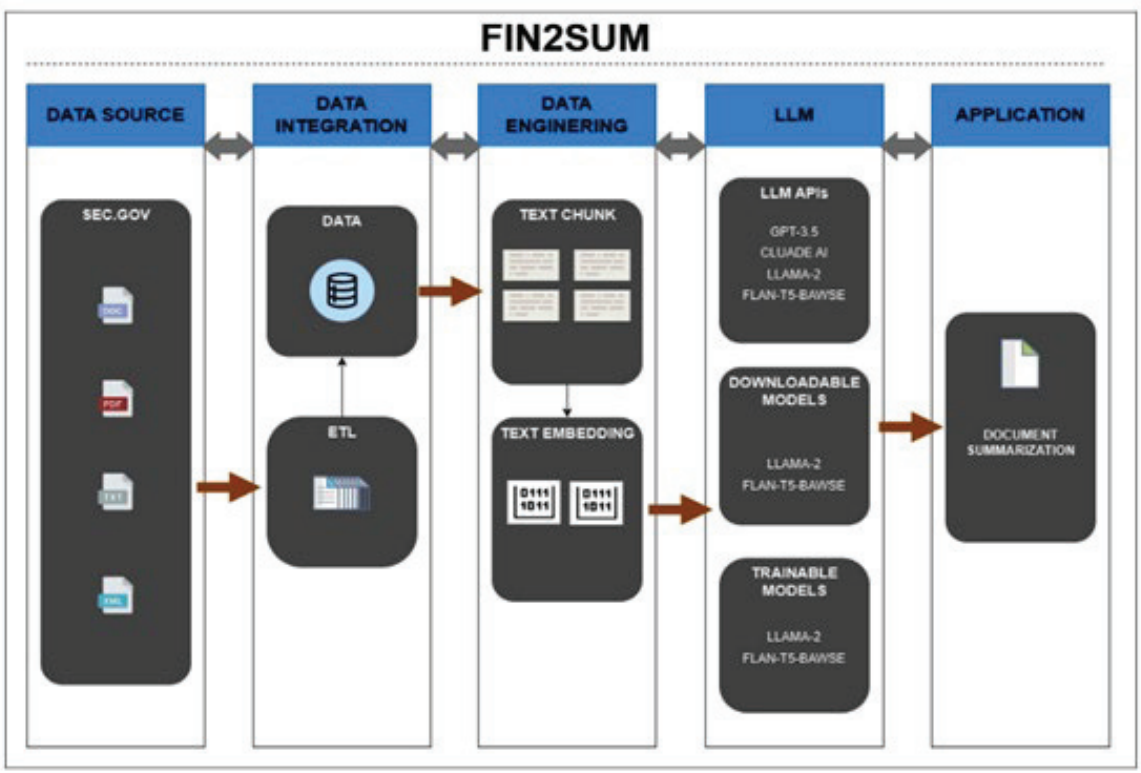


Fig. 3. FIN2SUM Architecture.

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User Experience and Security: The user-friendly FIN2SUM interface and strong data security measures ensure practicality and safety for real-world use.

C. Future work

Improving LLMs: Enhance language and context understanding in LLMs for more nuanced financial report analysis.

Expanding Dataset: Include a broader range of documents from various markets to test the model's adaptability.

UI Enhancements: Refine the user interface based on user feedback to benefit financial analysts.

Ethical AI and Regulation: Focus on ethical AI use and financial regulation compliance.

Predictive Analytics Integration: Combine FIN2SUM with predictive tools for better financial decision-making.

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